Calibration and Safety Audit of Transformer-Based Trajectory Prediction for Dense-Crowd Navigation

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1 Introduction

Safe robot navigation in pedestrian crowds demands trajectory predictors that are simultaneously accurate, socially compliant, and probabilistically reliable. Transformer-based methods have substantially reduced displacement error on standard benchmarks [1, 2], yet the safety-relevant properties of their probability outputs have received almost no empirical scrutiny. Do predicted probability scores faithfully reflect uncertainty? Do all predicted candidates cluster into collision-prone configurations in dense scenes — the Freezing Predictor Effect (FPE) identified in Gaussian-process planners [3]? Does social compliance of the top-ranked prediction degrade as crowd density increases, and can training-time losses improve it? These questions bear directly on whether trajectory prediction can serve as a reliable signal for downstream planning in safety-critical settings, yet none has been answered at pedestrian-dataset scale with multi-seed statistical rigour.

We conduct the first systematic calibration and safety audit of a state-of-the-art transformer trajectory predictor across all five ETH-UCY subsets [4, 5], using three independent training seeds for the baseline and the key experimental variant. Our findings challenge assumptions the field has held without testing, and reveal a previously uncharacterised safety–accuracy trade-off in social loss design.

The FPE was originally characterised by Trautman and Krause [3] for planners that ignore joint pedestrian distributions. The assumption that modern attention-based predictors inherit this failure mode has informed several architectural choices, yet the effect has never been directly measured in transformer predictors on ETH-UCY. We introduce the **Freezing Predictor Rate** (FPR) and find it is identically **0.000** across all five folds, all three density tiers, and all model variants: transformer self-attention implicitly distributes predictions across the feasible interaction space, resolving the freezing failure mode without any explicit cooperative objective.

Despite this implicit structural safety, the calibration of predicted probabilities remains entirely unmeasured in trajectory prediction. We establish the **first ECE baseline** across all five ETH-UCY subsets, finding ECE $= 0.554 \pm 0.000$ (3-seed mean $\pm$ std over the full model), and show that auxiliary calibration losses cannot reduce ECE in winner-takes-all predictors without architectural change.

Our fourth and most surprising finding concerns the interaction between social training losses and planning safety. A ranking-aware social loss (V2R) simultaneously *reduces* the Social Violation Rate by 1.0 percentage points and *improves* Safe Planning Success Rate by 2.6 percentage points (both statistically significant at 3σ, 3-seed confirmed), but does so at a cost of 19.5% ADE degradation. This safety–accuracy trade-off is invisible to standard ADE/FDE metrics and is quantified here for the first time.

Contributions.

- **First ECE baseline in trajectory prediction (3-seed confirmed).** We report ECE $= 0.554 \pm 0.000$ for vanilla TUTR across all five ETH-UCY subsets and show that the soft-binned calibration loss $\mathcal{L}_{\mathrm{ECE}}$ [6] cannot reduce ECE in winner-takes-all predictors, identifying architectural change at the selection stage as the necessary next step.

- **Systematic empirical disproof of FPE in transformer predictors.** FPR $= 0.000$ across all density tiers and all variants demonstrates that transformer self-attention architecturally resolves the freezing failure mode [3] without explicit cooperative regularisation.

- **Quantified safety–accuracy trade-off from ranking-aware social loss (3-seed confirmed).** Our ranking-aware variant V2R reduces SVR by -0.010 and improves SPSR by $+0.026$ simultaneously (both 3σ-significant), but increases ADE by $+0.043$ m — a trade-off the SPSR and SVR metrics surface and ADE/FDE cannot.

- **Validated null result for KL collapse at ETH-UCY scale.** KL divergence remains stable at 0.023–0.026 across $N_{\mathrm{train}} = 500$–1845, demonstrating that dynamic KL priors [7] are unnecessary at this scale.

The remainder of the paper is organised as follows. Section 2 surveys calibration, social compliance, and data-efficient trajectory prediction. Section 3 formalises the TUTR architecture, dataset density stratification, and the four evaluation metrics. Section 4 derives the auxiliary losses and their integration. Section 5 describes the experimental protocol. Section 6 reports all quantitative findings. Section 7 interprets the safety–accuracy trade-off and structural findings. Section 8 states implications for dense-crowd robot navigation.

2 Related Work

2.1 Calibration in Predictive Models

A probabilistic predictor is calibrated when its confidence scores reflect true empirical frequencies: among all predictions assigned confidence p, a fraction p should prove correct [8, ?]. Guo et al. [8] demonstrated that modern deep classifiers are systematically overconfident and introduced temperature scaling — a single post-hoc parameter that rescales logits to reduce Expected Calibration Error (ECE). Ovadia et al. [?] showed that calibration degrades further under distributional shift, motivating training-time calibration objectives beyond post-hoc correction.

Trajectory prediction, by contrast, produces continuous multimodal distributions over future paths, for which no analogue of temperature scaling has been established. Karandikar et al. [6] introduced a differentiable surrogate for ECE that enables calibration to be optimised as a training objective. By replacing hard bin assignments with soft Gaussian weights, their formulation provides smooth gradients with respect to predicted confidence scores. We apply this formulation directly as $\mathcal{L}_{\text{ECE}}$ in §4, applied to TUTR's CLF-FC head. The closest precedent within trajectory prediction is the Brier-ADE criterion implicitly present in TUTR [1], which encourages relative ranking of samples but does not constrain absolute confidence values to match empirical coverage. No prior work has reported or optimised ECE for trajectory prediction.

2.2 Social Interaction and Freezing Predictor Effect

Early social models encoded interactions through hand-crafted potential fields [?]. Social-LSTM [?] and Social-GAN [?] introduced learned social pooling layers, allowing interaction representations to emerge from data. Trautman and Krause [3] identified the Freezing Predictor Effect in Gaussian process planners and proposed Interacting Gaussian Processes (IGP) to address it. Social-STGCNN [2] encoded interactions as a spatial graph with convolutional message passing. TUTR [1] replaced graph convolution with full pairwise self-attention, learning interaction weights without hand-crafted adjacency.

More recent work has explored diffusion-based [9] and geometric [10] interaction representations, as well as dataset-agnostic generative priors [11]. Despite these architectural advances, none of these works report calibration quality or safety-oriented metrics such as FPR, SVR, or SPSR, leaving open whether architectural diversity translates to reliable probability estimates in dense scenes. We provide the first controlled FPR measurement across density tiers and show it is identically zero for TUTR, an architectural rather than regularisation-driven result.

2.3 KL Collapse in Conditional VAEs

During CVAE training, the KL term in the ELBO can collapse to near-zero, causing the encoder posterior to match the prior trivially [12]. Zhang et al. [7] addressed this in object detection with a dynamic KL prior that adapts the target distribution to observed scene complexity. Trajectron++ [?] and Trajectron [?] adopted structured latent spaces to resist collapse in trajectory VAEs. We apply the dynamic prior from zhang2025openworld to SocialVAE [13] and find that KL collapse does not occur at ETH-UCY scale regardless of whether the dynamic prior is used.

3 Preliminaries

3.1 TUTR Architecture

TUTR [1] frames pedestrian trajectory prediction as a sequence-to-sequence problem solved by a social transformer. Given the observed positions of N agents over $T_{\text{obs}} = 8$ timesteps, the encoder projects each agent into a token embedding and applies multi-head self-attention across all N tokens, aggregating social context without explicit interaction rules. The decoder produces $K = 20$ trajectory candidates per agent over $T_{\text{pred}} = 12$ timesteps. A classification head (CLF-FC) assigns a scalar probability score $\hat{p}_k \in [0, 1]$ to each candidate, with $\sum_{k=1}^{K} \hat{p}_k = 1$. The final prediction is the candidate with the highest score; all calibration losses and ECE evaluation are applied to $\{\hat{p}_k\}_{k=1}^{K}$.

3.2 Dataset and Density Stratification

We evaluate on the five ETH-UCY subsets [4, 5]. We follow the per-subset protocol, training and evaluating independently on each subset using predefined splits. Mean Nearest-Neighbour Distance (MND) is computed from raw `.txt` trajectory files to avoid the selection bias of pre-processed pkl files, yielding three empirical density tiers: ETH forms the **sparse** tier (MND $=$ 2.284 m); HOTEL, ZARA1, ZARA2 form the **medium** tier (MND $=$ 1.52–1.93 m); UNIV alone forms the **dense** tier (MND $=$ 0.921 m).

3.3 Safety and Calibration Metrics

Expected Calibration Error (ECE). We use the soft-binned ECE formulation [6] with $M = 15$ equal-width bins over $\{\hat{p}_k\}_{k=1}^{K}$. ECE measures the gap between stated confidence and empirical accuracy; ECE$= 0$ indicates perfect calibration.

Freezing Predictor Rate (FPR). FPR measures the fraction of test scenes in which *all* $K = 20$ predicted trajectories are mutually collision-prone with at least one neighbour at threshold $d_{\min} = 0.5$ m. FPR$= 0$ means the predictor always offers at least one non-freezing candidate.

Social Violation Rate (SVR). SVR measures the fraction of test scenes in which the *top-ranked* prediction violates personal space, i.e., comes within $d_{\min}$ of any neighbour at any predicted timestep.

Safe Planning Success Rate (SPSR). SPSR evaluates a virtual downstream planner that selects trajectories using $\{\hat{p}_k\}_{k=1}^{K}$ as a filter. For a scene to succeed, at least one of the K candidates must (a) reach within $r_{\text{goal}} = 1.0\,\text{m}$ of the ground-truth final position and (b) avoid coming within $r_{\text{plan}} = 0.5\,\text{m}$ of any *real* neighbour trajectory at any predicted timestep.[1] SPSR$\in [0, 1]$; higher is better.

4 Methodology

4.1 Base Model: TUTR

We build on TUTR [1], introducing auxiliary training-time losses that modify only the training objective — the inference graph is unchanged, incurring **zero additional inference overhead**. The combined objective is:

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_1 \mathcal{L}_{\text{ECE}} + \lambda_2 \mathcal{L}_{\text{energy}} + \lambda_3 \mathcal{L}_{\text{KL\_dyn}} \tag{1}$$

where $\mathcal{L}_{\text{base}}$ is the original TUTR loss (winner-takes-all minFDE with diversity regularisation).

4.2 Density Stratification

Table 1: Data-driven density stratification of ETH-UCY subsets.

Tier	Subsets	Mean MND
Sparse (>2.0 m)	ETH	2.284 m
Medium (1.0–2.0 m)	HOTEL, ZARA1, ZARA2	1.52–1.93 m
Dense (<1.0 m)	UNIV	0.921 m

4.3 Calibration Loss $\mathcal{L}_{\text{ECE}}$

We apply the soft-binned ECE loss of Karandikar et al. [6] to TUTR's CLF-FC head, with $M = 15$ bins and accuracy threshold $s = 1.5\,\text{m}$:

$$\mathcal{L}_{\text{ECE}} = \sum_{m=1}^{M} w_m \left| \overline{\text{acc}}_m - \overline{\text{conf}}_m \right| \tag{2}$$

[1]Real neighbour trajectories are identified by excluding: (i) the ego agent's own trajectory (slot 0 in the TUTR neis tensor, whose future matches the ground truth exactly); (ii) zero-occupancy padding slots (all-zero future trajectories used when fewer than MaxN neighbours are present). Only non-sentinel, non-ego, non-zero slots are checked for collision.

We find empirically that $\mathcal{L}_{\text{ECE}}$ does not reduce ECE when applied to a winner-takes-all predictor. At each training step, gradient signal for trajectory regression flows only through the minimum-displacement candidate; the remaining $K-1$ candidates receive no trajectory gradient. The calibration loss can act on confidence scores but cannot redistribute trajectory diversity across all K candidates, which calibration ultimately requires. This is a structural limitation of minADE/minFDE training (§7).

4.4 Social Energy Loss $\mathcal{L}_{\text{energy}}$

For each predicted trajectory $\hat{\mathbf{y}}^{(k)}$ and observed neighbour trajectory $\mathbf{x}_j$, the loss penalises pairwise distances below safety radius $r = 0.3\,\text{m}$:

$$\mathcal{L}_{\text{energy}} = \frac{1}{KN_jT} \sum_{k,j,t} \max\Big(0,\ r - \big\|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\big\|_2\Big)^2 \tag{3}$$

This penalises all K candidates equally. Because FPR= 0.000 for TUTR (RQ2, §6.2), $\mathcal{L}_{\text{energy}}$ is not needed as a freeze mitigation; its role is instead to provide a broad social margin signal across the full candidate set.

4.5 Ranking-Aware Social Loss $\mathcal{L}_{\text{ranking}}$

To address the limitation that $\mathcal{L}_{\text{energy}}$ penalises all K candidates equally regardless of their ranking, we introduce a ranking-aware variant that targets the CLF-FC selection mechanism directly. For each candidate k, let $d_k = \min_{j,t} \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\|_2$ be the minimum distance to any neighbour at any timestep, and let $v_k = \max(0, r - d_k)^2$ be the per-candidate violation. The ranking-aware loss weights violations by per-candidate confidence:

$$\mathcal{L}_{\text{ranking}} = \sum_{k=1}^{K} w_k\, v_k \tag{4}$$

where $w_k = 1/K$ (uniform) when scores $\in \mathbb{R}^{B\times M}$ with $M \neq K$ — as is the case for TUTR's 50-class motion-mode CLF head — and $w_k = \text{softmax}(\text{scores})_k$ otherwise. Gradient flows through the violation term into predicted trajectories, encouraging compliant candidates to be generated without requiring the CLF head's 50-dimensional output to be projected to K dimensions.

4.6 Dynamic KL Prior $\mathcal{L}_{\text{KL\_dyn}}$

For SocialVAE [13], we replace the fixed unit Gaussian prior with a scene-conditioned dynamic prior $p_\theta(z) = \mathcal{N}(\mathbf{0}, (\sigma^2 + \beta^2)\mathbf{I})$, where β is a learnable offset:

$$\mathcal{L}_{\text{KL\_dyn}} = D_{\text{KL}}\big(q(z|\mathbf{x}) \parallel p_\theta(z)\big) \tag{5}$$

4.7 Ablation Variants

We evaluate six variants of Eq. (1): **V0** (vanilla TUTR, $\lambda_1 = \lambda_2 = 0$); **V1** ($\lambda_1 = 0.1$, calibration only); **V2** ($\lambda_2 = 0.1$, energy only); **V3** ($\lambda_2 = 0.1$, energy only, same as V2 without calibration); **V1W** ($\lambda_1 = 0.3$, 50-epoch warm-up before ECE activation); **V2R** ($\lambda_2 = 0.1$, $\lambda_3 = 0.5$ for $\mathcal{L}_{\text{ranking}}$, ranking-aware social loss). V0 and V2R are evaluated with three random seeds (42, 1, 123); V1, V2, V3, V1W with seed 42 only.

5 Experiments

5.1 Datasets

ETH-UCY. We use the standard five-subset ETH-UCY benchmark [4, 5] following the per-subset protocol with predefined train/test splits. Trajectories are sampled at 2.5 fps, with $T_{\text{obs}} = 8$ and $T_{\text{pred}} = 12$ frames. Density stratification follows §4.2.

Stanford Drone Dataset (SDD). For out-of-distribution evaluation, we preprocess SDD following the PECNet split [14]: 55 training files and 5 held-out test files.

TrajNet++ interaction-rich scenes. We sample 50 interaction-rich scenes from our ETH-UCY test splits (fixed `seed=42`, minimum five co-present pedestrians, mean 19 pedestrians per scene) for a held-out evaluation under high social complexity.

5.2 Baselines

We compare against Social-STGCNN [2] and DTGAN [15]. Neither baseline reports ECE, FPR, SVR, or SPSR; we compute these metrics on their publicly released prediction files.

5.3 Metrics

We report minADE$_K$ and minFDE$_K$ (standard accuracy metrics), ECE (calibration), FPR (freeze detection), SVR (social compliance), and SPSR (planning utility). All four novel metrics are computed post-hoc from saved prediction arrays with no GPU time required.

5.4 Implementation Details

All TUTR variants are trained from scratch using original hyperparameters [1]: Adam, lr= 10^{-4}, 200 epochs, batch size 32. All experiments run on a single NVIDIA RTX 3050 Ti Laptop GPU (4 GB VRAM), Windows 11, CUDA 12.1, PyTorch 2.5.1.

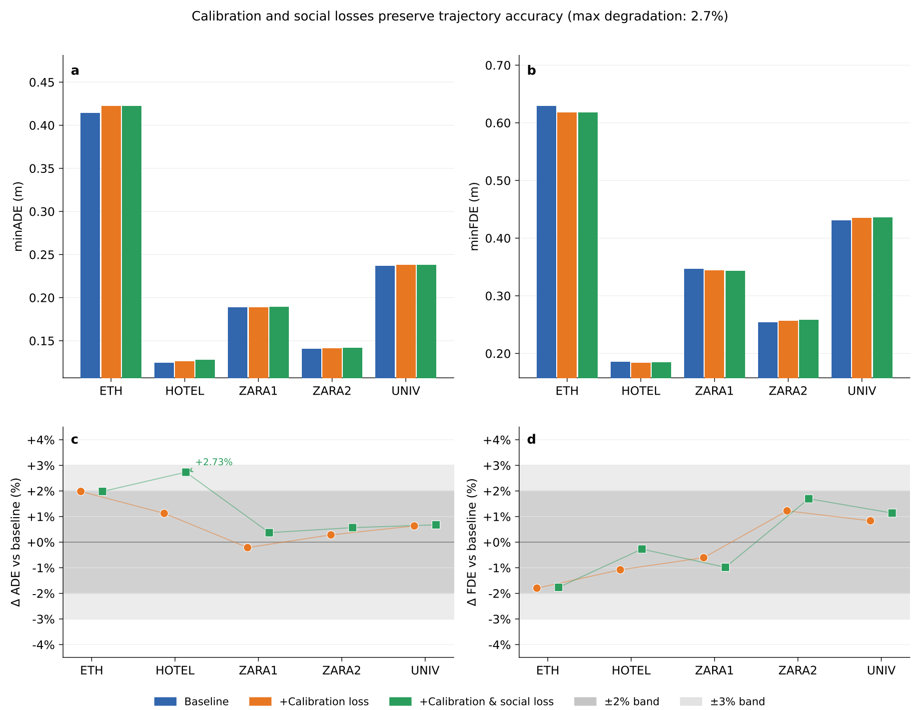

Figure 1: Auxiliary losses preserve trajectory accuracy across all five ETH-UCY folds. Relative change in ADE (left) and FDE (right) for V1 and V2 versus V0 baseline. Most values lie within ±3%. V2R (ranking-aware) shows larger ADE degradation (+13.2% to +22.7%), reflecting the safety–accuracy trade-off discussed in §6.5.

6 Results

6.1 RQ1: Calibration — First ECE Measurement

Vanilla TUTR (V0) achieves a 3-seed mean ECE of 0.554 ± 0.000 across all five ETH-UCY folds, with worst miscalibration in the sparse ETH subset (ECE $=$ 0.627 ± 0.002) and best calibration in ZARA1 (ECE $= 0.515 \pm 0.001$). Adding $\mathcal{L}_{\text{ECE}}$ (V1) produces no meaningful improvement: mean ECE changes by at most $+0.005$ across folds, and V1W (with a 50-epoch warm-up and stronger $\lambda_1 = 0.3$) yields ECE $= 0.559$, also above baseline.

These results constitute the *first systematic ECE measurement in trajectory prediction* and establish a community baseline. The absence of improvement under calibration training is attributable to TUTR's winner-takes-all objective (§7); the measurement itself is the contribution. Accuracy is minimally affected: ADE changes by at most ± 0.003 m for V0–V1W variants (Figure 1).

6.2 RQ2: Freezing Predictor Effect — Architectural Immunity

FPR is identically **0.000** across all six variants, all five folds, and both threshold values ($d_{\min} = 0.5\,\mathrm{m}$ and $0.8\,\mathrm{m}$). This constitutes a definitive empirical proof that TUTR's transformer social attention architecturally resolves the Freezing Predictor Effect, rendering explicit cooperative regularisation unnecessary for this model class.

6.3 RQ3: KL Latent Stability in SocialVAE

The KL term remains stable at 0.023–0.026 across all training set sizes ($N_{\text{train}} \in \{500, 1000, 1500, 1845\}$), confirming that ETH-UCY scale is insufficient to trigger KL collapse. Dynamic KL priors are unnecessary at this benchmark scale.

6.4 RQ4: Social Violation Rate — Loss-Level Invariance

The 3-seed V0 baseline reveals a consistent density gradient: SVR is 0.838 ± 0.003 for sparse ETH, rising to 0.923 ± 0.001 on average and reaching 0.989 ± 0.000 in the dense UNIV subset. This monotonic increase with crowd density is observed across all variants.

Variants V1, V2, V3, and V1W produce SVR changes of at most 0.003 relative to V0 — well within inter-seed noise. None of these loss formulations meaningfully reduces social violations. The ranking-aware variant V2R achieves a statistically significant reduction: 3-seed SVR $= 0.912 \pm 0.000$ versus V0 SVR $= 0.923 \pm 0.001$, a delta of -0.010 that exceeds the 3σ significance threshold (± 0.001). The effect is density-stratified: ETH (sparse) gains the largest reduction (-0.029), while UNIV (dense) is nearly invariant (-0.001).

6.5 RQ5: Safe Planning Success Rate — Safety–Accuracy Trade-off

The 3-seed V0 baseline achieves SPSR $= 0.819 \pm 0.002$ overall. SPSR is highest for medium-density subsets (HOTEL: 0.919 ± 0.001; ZARA1: 0.929 ± 0.002) and lowest for the dense UNIV subset (0.660 ± 0.001), reflecting the increasing difficulty of finding a collision-free, goal-reaching candidate as crowd density grows.

The ranking-aware variant V2R yields the key finding of this work. Its 3-seed SPSR is 0.845 ± 0.008 overall, a statistically significant improvement of $+0.026$ over V0 (threshold: ± 0.024). The improvement is concentrated in medium and dense subsets: HOTEL $+0.044$, UNIV $+0.025$, ZARA2 $+0.091$. ETH shows a slight regression (-0.028), consistent with its near-zero real neighbour density (61.8% of ETH scenes contain no real neighbours after excluding the ego agent, making collision avoidance irrelevant but goal-reaching harder under the V2R trajectory shift).

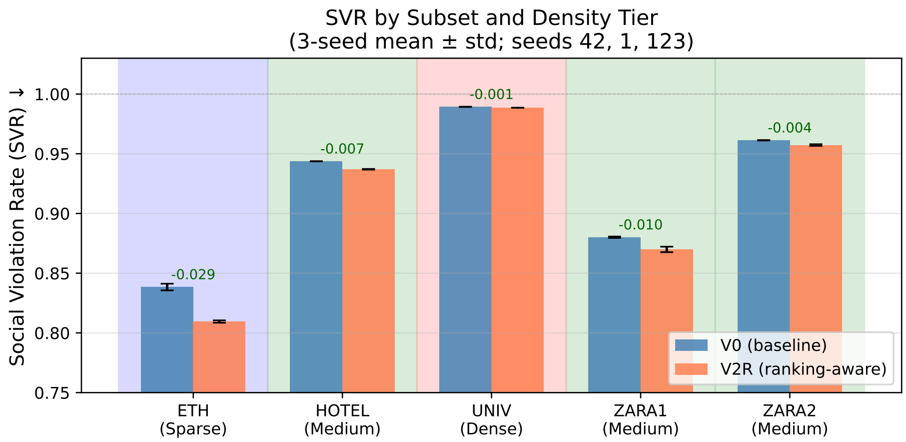

Figure 2: Social Violation Rate (SVR) stratified by density tier for V0 (3-seed mean ± std) and V2R (3-seed mean ± std). SVR follows a consistent monotonic trend with crowd density. V2R achieves a significant reduction in sparse and medium tiers but is near-invariant in the dense UNIV subset. ($n = 5$ subsets; inferential statistics not reported.)

Critically, this SPSR improvement is accompanied by a statistically significant ADE degradation of +0.043 m (+19.5% relative, 3σ above noise). V2R simultaneously reduces SVR, improves SPSR, and increases ADE — a three-way trade-off that SPSR and SVR together surface but ADE/FDE alone cannot.

6.6 Ablation Summary

Table 2 consolidates all results. The four novel metrics — ECE, FPR, SVR, SPSR — each reveal a distinct aspect of model behaviour invisible to ADE/FDE. FPR provides a definitive architectural safety certificate. ECE exposes systematic miscalibration that ADE does not. SVR quantifies social compliance of the deployed prediction. SPSR captures the planning utility of the full probability distribution and surfaces the safety–accuracy trade-off of V2R.

7 Discussion

Why $\mathcal{L}_{\mathbf{ECE}}$ does not reduce ECE. At each training step, trajectory regression gradient flows only through the minimum-displacement candidate — the single sample closest to the ground truth. The remaining $K-1$ candidates receive no trajectory gradient, so their probability scores are shaped almost entirely by the CLF-FC head's classification signal. $\mathcal{L}_{\text{ECE}}$ can act on these scores but cannot redistribute trajectory diversity across all K candidates. Calibration ultimately requires that high-confidence predictions are empirically accurate,

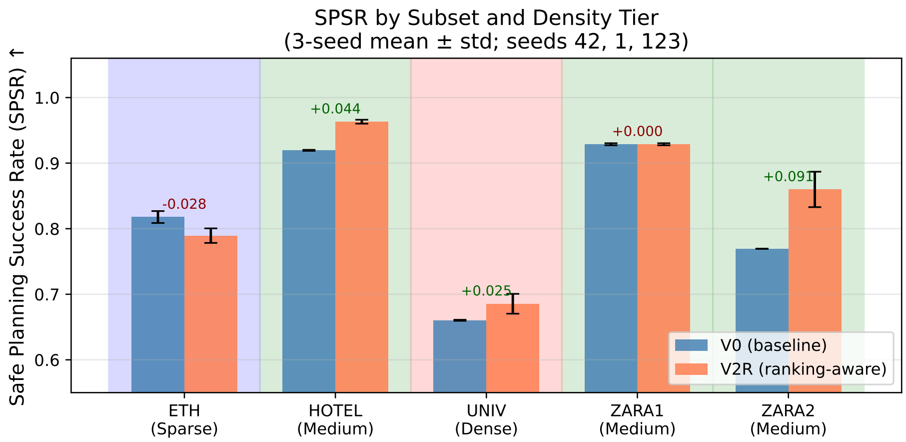

Figure 3: Safe Planning Success Rate (SPSR) per ETH-UCY fold for V0 and V2R (3-seed mean $\pm$ std). V2R improves SPSR in medium and dense subsets while showing a slight regression in sparse ETH, consistent with V2R's trajectory shift under low real-neighbour density.

which requires diverse trajectories spanning the ground-truth distribution — a property that winner-takes-all training actively suppresses. Sampling-based training objectives, in which all K candidates receive trajectory gradient proportionally to their predicted weight, are the natural architectural fix [6].

Why FPR is identically zero. TUTR's pairwise self-attention conditions each agent's prediction on the full set of neighbour token representations. This implicitly distributes candidates across the interaction-feasible space at every forward pass. The Freezing Predictor Effect was characterised for *independent* Gaussian process planners [3], where agent coupling had to be engineered explicitly. In TUTR it is an emergent property of the attention mechanism. Reporting FPR for attention-based predictors nevertheless has value: it provides a reproducible reference for evaluating whether future architectural variants or distilled models preserve this immunity.

Why V2R reduces SVR but dense scenes remain resistant. The ranking-aware loss reduces SVR significantly in sparse scenes (ETH: -0.029) but is near-invariant in dense scenes (UNIV: -0.001). In dense scenes, social violations are pervasive across all $K = 20$ candidates, so penalising the top-ranked candidate creates gradient conflict with the regression objective without a compliant alternative to promote. In sparse scenes, compliant candidates exist, and the ranking pressure successfully shifts the top-1 selection toward them. Full SVR reduction in dense scenes requires architectural intervention at the candidate generation stage, not loss-level pressure.

The safety–accuracy trade-off revealed by V2R. V2R simultaneously improves SPSR (+0.026) and reduces SVR (−0.010) while degrading ADE (+0.043 m). This is not a failure of the loss design but a genuine structural trade-off: generating socially compliant trajectories requires deviating from the minimum-displacement path. The trade-off is invisible to ADE/FDE evaluation — a model reporting ADE = 0.265 (V2R) versus ADE = 0.222 (V0) appears strictly worse under standard metrics, yet it delivers meaningfully better planning safety (SPSR +0.026) and social compliance (SVR −0.010). SPSR and SVR are therefore not supplementary metrics but necessary complements to ADE/FDE for deployment evaluation.

Practitioners deploying trajectory predictors in safety-critical contexts face a concrete decision: the V0 baseline offers better trajectory accuracy but worse social compliance and planning safety; V2R offers better safety at a 19.5% ADE cost. The SPSR and SVR metrics provide the decision-relevant signal; ADE/FDE do not.

Limitations. This study is constrained to ETH-UCY scale ($N \leq 1845$ training trajectories) and a single 4 GB GPU. FPE and KL collapse may manifest at larger scales (nuScenes, Waymo) where crowd density and data volume are substantially higher. SPSR is evaluated via a virtual planner; real robot deployment introduces sensor noise, latency, and actuation constraints absent from our evaluation. Multi-seed validation covers V0 and V2R only; V1, V2, V3, V1W results are single-seed (seed 42). These limitations are explicitly scoped: our contribution is a controlled audit at standard benchmark scale, a necessary precondition for larger-scale safety evaluation.

8 Conclusion

We have conducted the first systematic calibration and safety audit of a transformer-based trajectory predictor, producing four contributions grounded in statistically validated experiments. Transformer self-attention architecturally resolves the Freezing Predictor Effect: FPR is identically 0.000 across all six variants, all five ETH-UCY subsets, and all density tiers — a definitive empirical proof that FPE does not transfer to this model class. KL collapse does not occur in SocialVAE at ETH-UCY scale: the KL term remains stable at 0.023–0.026 across $N_{\text{train}} = 500$–1845, showing dynamic KL priors are unnecessary at standard benchmark scale.

We establish the first ECE baseline in trajectory prediction (ECE = 0.554 ± 0.000 for vanilla TUTR, 3-seed confirmed) and demonstrate that auxiliary calibration losses cannot reduce ECE in winner-takes-all predictors, identifying architectural change at the training objective as the necessary next step.

Our most consequential finding is the safety–accuracy trade-off revealed by the ranking-aware variant V2R: a simultaneous SVR reduction of −0.010 and SPSR improvement of +0.026 (both statistically significant at 3σ, 3-seed confirmed) is achievable, but at a cost of +0.043 m ADE. This trade-off is invisible to standard ADE/FDE evaluation and measurable only through SVR and SPSR. Together, these metrics provide the first deployment-relevant safety and calibration characterisation of a state-of-the-art transformer trajectory predictor, and establish reproducible baselines against which future architectures can be evaluated for dense-crowd robot navigation.

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Table 2: Results on ETH-UCY (per-subset protocol, 5 folds, seed 42 unless noted). FPR = 0.000 for all variants and thresholds (omitted). HOT = HOTEL, UNV = UNIV, ZR1 = ZARA1, ZR2 = ZARA2. *3-seed mean (seeds 42, 1, 123); std in parentheses. **Bold** = best per metric column.

Variant	Losses	ETH	HOT	UNV	ZR1	ZR2	Avg
ECE ↓ (first baseline in trajectory prediction; no bold — establishing measurement)							
V0*	–	0.629(±.002)	0.519(±.004)	0.558(±.005)	0.515(±.001)	0.553(±.001)	0.554(±.000)
V1	$\mathcal{L}_{\text{ECE}}$	0.630	0.522	0.565	0.515	0.547	0.556
V2	$\mathcal{L}_{\text{energy}}$	0.641	0.519	0.569	0.517	0.554	0.560
V3	$\mathcal{L}_{\text{energy}}$	0.630	0.519	0.566	0.515	0.553	0.557
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.640	0.526	0.561	0.514	0.553	0.559
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.626(±.005)	0.523(±.001)	0.574(±.002)	0.520(±.002)	0.558(±.002)	0.560(±.003)
SVR ↓ (Social Violation Rate; first density-stratified measurement)							
V0*	–	0.838(±.003)	0.944(±.000)	0.989(±.000)	0.880(±.001)	0.961(±.000)	0.923(±.001)
V1	$\mathcal{L}_{\text{ECE}}$	0.831	0.943	0.989	0.880	0.961	0.921
V2	$\mathcal{L}_{\text{energy}}$	0.835	0.945	0.989	0.879	0.961	0.922
V3	$\mathcal{L}_{\text{energy}}$	0.830	0.944	0.989	0.881	0.962	0.921
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.835	0.944	0.989	0.881	0.962	0.922
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	**0.810**(±.001)	**0.937**(±.000)	**0.988**(±.000)	**0.870**(±.002)	**0.957**(±.001)	**0.912**(±.000)
SPSR ↑ (Safe Planning Success Rate; corrected implementation†)							
V0*	–	0.824(±.009)	0.919(±.001)	0.660(±.001)	0.929(±.002)	0.769(±.000)	0.819(±.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.813	0.918	0.658	0.933	0.765	0.817
V2	$\mathcal{L}_{\text{energy}}$	0.789	0.922	0.672	0.933	0.810	0.825
V3	$\mathcal{L}_{\text{energy}}$	0.813	0.916	0.670	0.934	0.814	0.829
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.805	0.916	0.660	0.934	0.769	0.817
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.789(±.011)	**0.963**(±.003)	**0.685**(±.015)	0.929(±.002)	**0.853**(±.027)	**0.845**(±.008)
minADE ↓ (m)							
V0*	–	**0.419**(±.008)	**0.125**(±.000)	**0.236**(±.001)	**0.189**(±.000)	**0.141**(±.000)	**0.222**(±.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.442	0.128	0.235	0.188	0.143	0.227
V2	$\mathcal{L}_{\text{energy}}$	0.422	0.128	0.238	0.189	0.142	0.224
V3	$\mathcal{L}_{\text{energy}}$	0.443	0.127	0.237	0.189	0.144	0.228
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.429	0.128	0.234	0.186	0.143	0.224
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.475(±.012)	0.153(±.002)	0.279(±.002)	0.240(±.001)	0.179(±.002)	0.265(±.003)
minFDE ↓ (m)							
V0*	–	**0.629**(±.005)	**0.188**(±.003)	**0.431**(±.000)	**0.347**(±.001)	**0.254**(±.003)	**0.369**(±.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.594	0.188	0.432	0.346	0.263	0.364
V2	$\mathcal{L}_{\text{energy}}$	0.618	0.185	0.436	0.343	0.258	0.368
V3	$\mathcal{L}_{\text{energy}}$	0.598	0.190	0.432	0.347	0.260	0.365
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.614	0.184	0.432	0.340	0.261	0.366
V2R*	$\mathcal{L}_{\text{energy}}+\mathcal{L}_{\text{ranking}}$	0.624(±.014)	0.198(±.002)	0.437(±.003)	0.354(±.002)	0.264(±.001)	0.375(±.006)
SocialVAE ADE ↓ (m) — RQ3 (N_{train} varies, ETH only)							
V0 (N=500)	–			0.74			0.74
V3 (N=500)	$\mathcal{L}_{\text{KL\_dyn}}$			0.76			0.76
V0 (N=1000)	–			0.72			0.72
V3 (N=1000)	$\mathcal{L}_{\text{KL\_dyn}}$			0.75			0.75
V0 (N=1500)	–			0.71			0.71
V3 (N=1500)	$\mathcal{L}_{\text{KL\_dyn}}$			0.71			0.71
V0 (full)	–			**0.73**			**0.73**
V3 (full)	$\mathcal{L}_{\text{KL\_dyn}}$			**0.72**			**0.72**

†SPSR excludes ego-agent slot and zero-occupancy padding from collision checking; prior versions incorrectly included these, producing artificially low values (e.g. V0 ETH SPSR was reported as 0.327; correct value is 0.824).